# Angel Peredo

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## Education

**University of Texas Rio Grande Valley (UTRGV)** Edinburg, TX
Ph.D. in Computer Science Aug 2025 – Present
Research: Federated Learning, Generalization in Non-IID Systems
Advisor: Dr. Sergei Chuprov
GPA: 4.0/4.0

**University of Texas Rio Grande Valley (UTRGV)** Edinburg, TX
M.S. in Computer Science Aug 2023 – May 2025
Thesis: Drug-Target Binding Affinity Prediction with Graph-Based Models
GPA: 3.82/4.0
Awards: Outstanding Thesis Award, Presidential Research Fellowship

**University of Texas Rio Grande Valley (UTRGV)** Edinburg, TX
B.S. in Computer Science Jan 2019 – May 2023
GPA: 3.37/4.0

## Research Experience

**Graduate Research Assistant (Ph.D.)** Aug 2025 – Present
University of Texas Rio Grande Valley

- Developed personalized Federated Learning aggregation methods to improve generalization under highly non-IID data distributions, evaluated on large-scale simulated client networks using HPC infrastructure.
- Designed a Vision-Language Model (VLM) and computer vision pipeline for flood detection and depth estimation, integrating multimodal data for real-world deployment with the Texas Department of Transportation (TxDOT).
- Conducted large-scale training and simulation of federated systems using parallelized workloads on HPC clusters managed with Slurm.

**Graduate Research Assistant (M.S.)** Aug 2023 – May 2025
Machine Intelligence Lab, UTRGV

- Developed a target-aware regression framework for drug-target binding affinity prediction using Equivariant Graph Neural Networks and Graph Transformers, earning the Outstanding Thesis Award.
- Fine-tuned Transformer-based models (BERT) for molecular toxicity prediction using RDKit and PyMOL for structural feature extraction.

- Implemented offline Reinforcement Learning algorithms for safe and efficient smart grid control in simulated environments.
- Managed GPU cluster resources and mentored undergraduate researchers in machine learning methodologies.

**Undergraduate Research Assistant** Jan 2023 – May 2023
University of Texas Rio Grande Valley

- Developed a deep Reinforcement Learning system for motion imitation using IsaacGym.
- Trained simulated humanoid agents to replicate human tennis motions from motion capture (MOCAP) data.

## Publications & Presentations

- **Peredo, A.**, Chuprov, S. (2026). Improving Generalization with Harmonic Aggregation in Personalized Federated Learning. *IEEE Consumer Communications & Networking Conference (CCNC).*
- **Peredo, A.**, Lugo, H., Narcia-Macias, C., et al. Offline Reinforcement Learning Approaches for Safe and Effective Smart Grid Control. *International Congress on Information and Communication Technology (ICICT).*
- **Peredo, A.**, Aboagye-Otchere, K. B., Bhatta, S. Reinforcement Learning-Based Recommender System for MOOCs. *Emerging Researchers National (ERN) Conference.*

## Technical Skills

**Core Machine Learning:** PyTorch, Federated Learning, Reinforcement Learning, Graph Neural Networks (GNNs), Transformers, Vision-Language Models
**Programming:** Python, C++, SQL
**Scientific Tools:** RDKit, PyMOL, AutoDock Vina
**Systems:** High-Performance Computing (HPC), Slurm, Linux/Unix

## Honors & Awards

- Outstanding Thesis Award, UTRGV Computer Science Department (2025)
- Presidential Research Fellowship, UTRGV (2023–2025)
- MIT Reality Hack Participant (2025)